TRADITIONAL

TEXT CLASSIFICATION

Document Classification: Motivation

- News article classification
- Automatic email filtering
- Webpage classification
- Word sense disambiguation
- … …

Introduction

- Purpose: classification of natural language texts into a set of predefined labels.

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spam? Or legitimate?

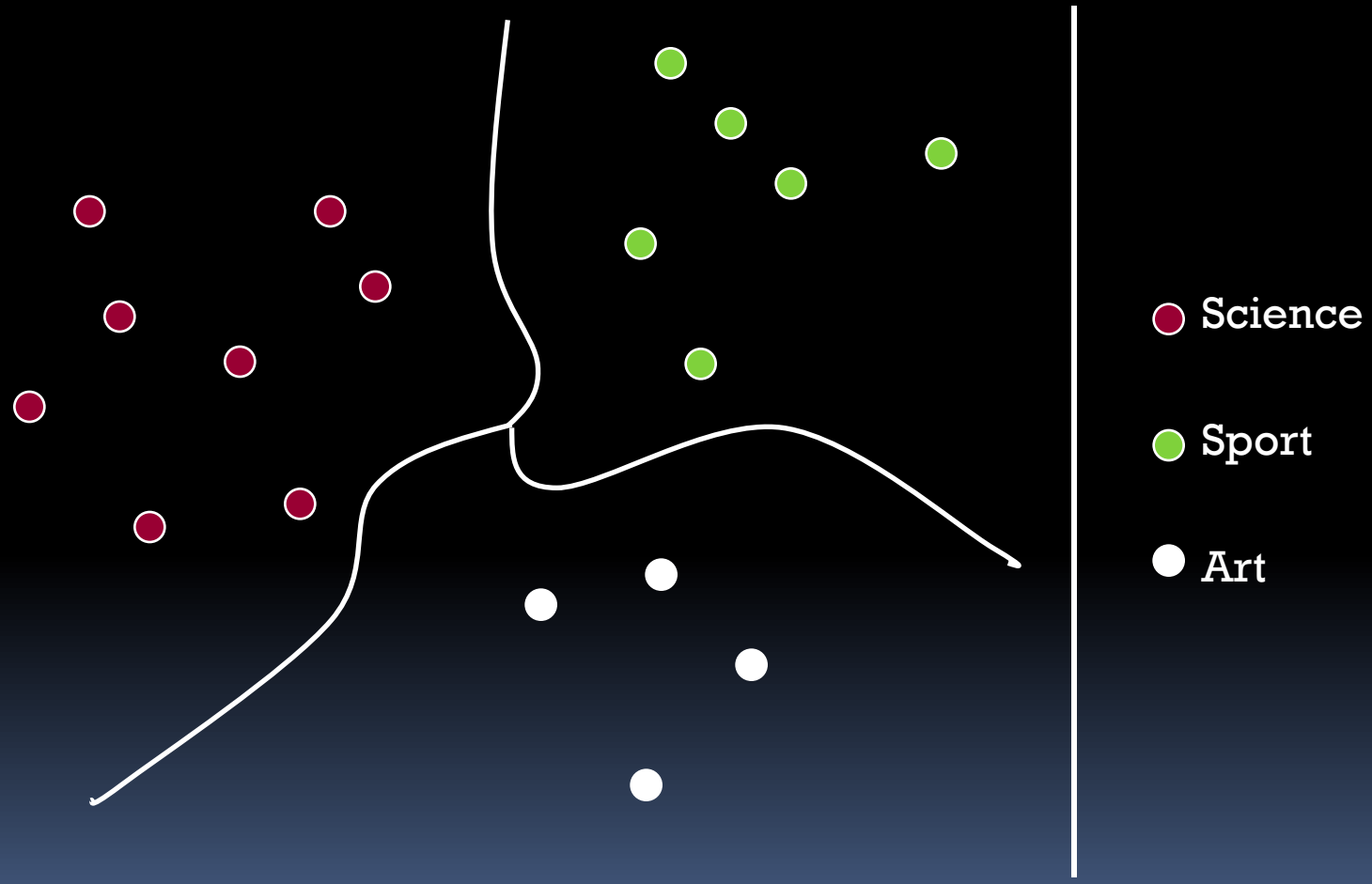
Main Uses

- Indexing (e.g. Libraries)
- Organization
  - News articles (Reuters, GoogleNews)
  - Classified (Craigslist)
  - Webpages (Yahoo Directory)
- Filtering
  - News Feed
  - Spam

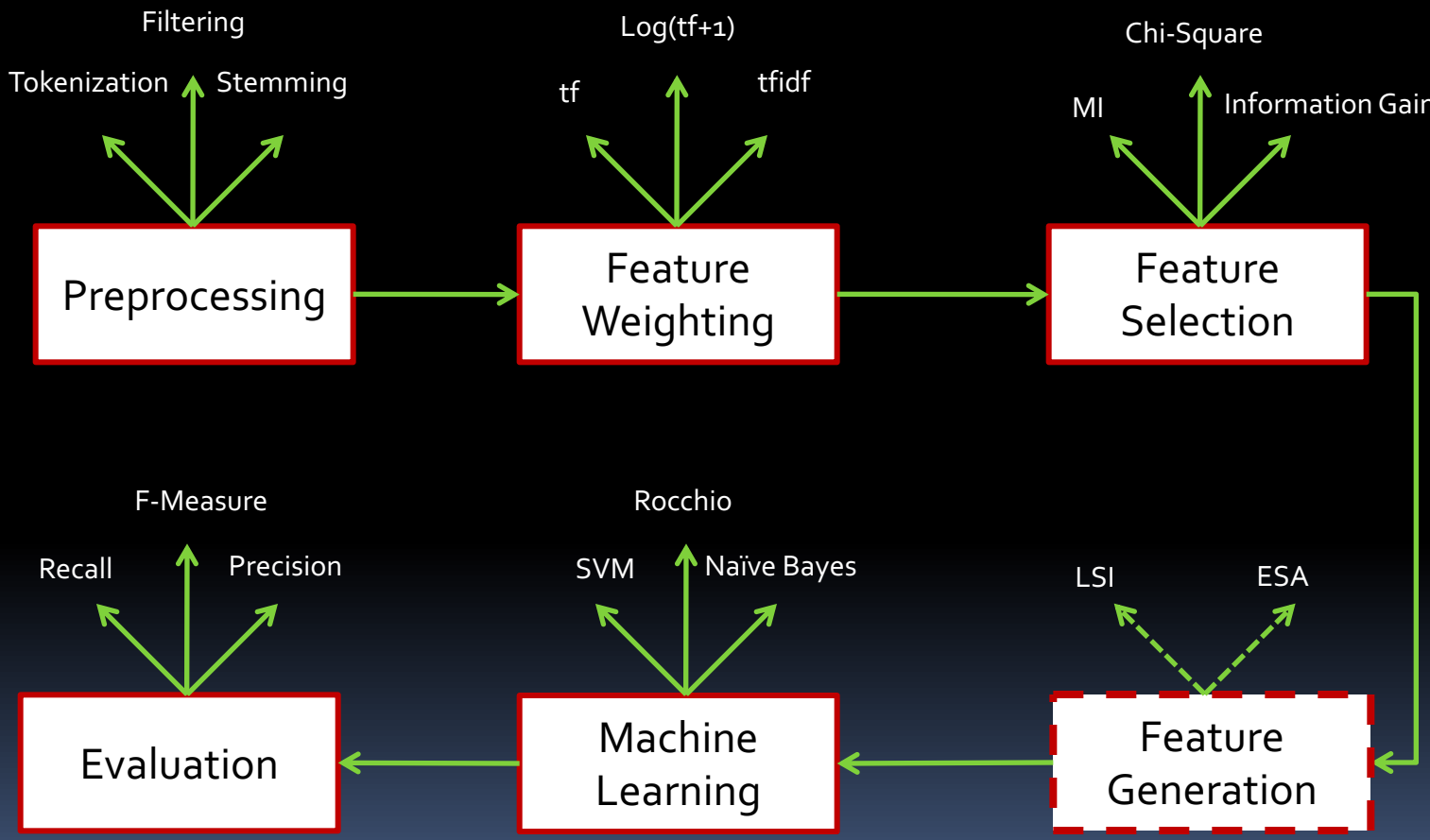
Other Uses

- Author Identification
- Genre Detection
- Language Identification
- Sentiment Classification
  - Market Analysis (Reuters)

Illustration of Text Classification



Anatomy of a Text Categorizer



Classification types

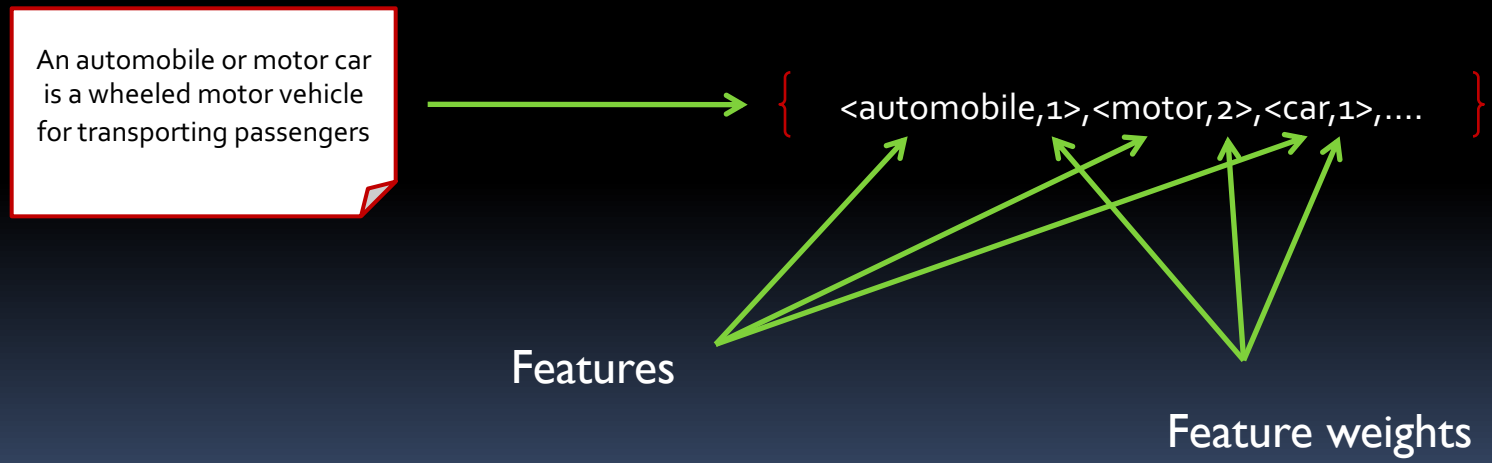
- Document Membership
  - Single Label
  - Multiple Labels
  - Binary
- Hard vs Ranking Classifiers
  - Hard = Decisive!
  - Ranking = Probabilistic

Classification Methods

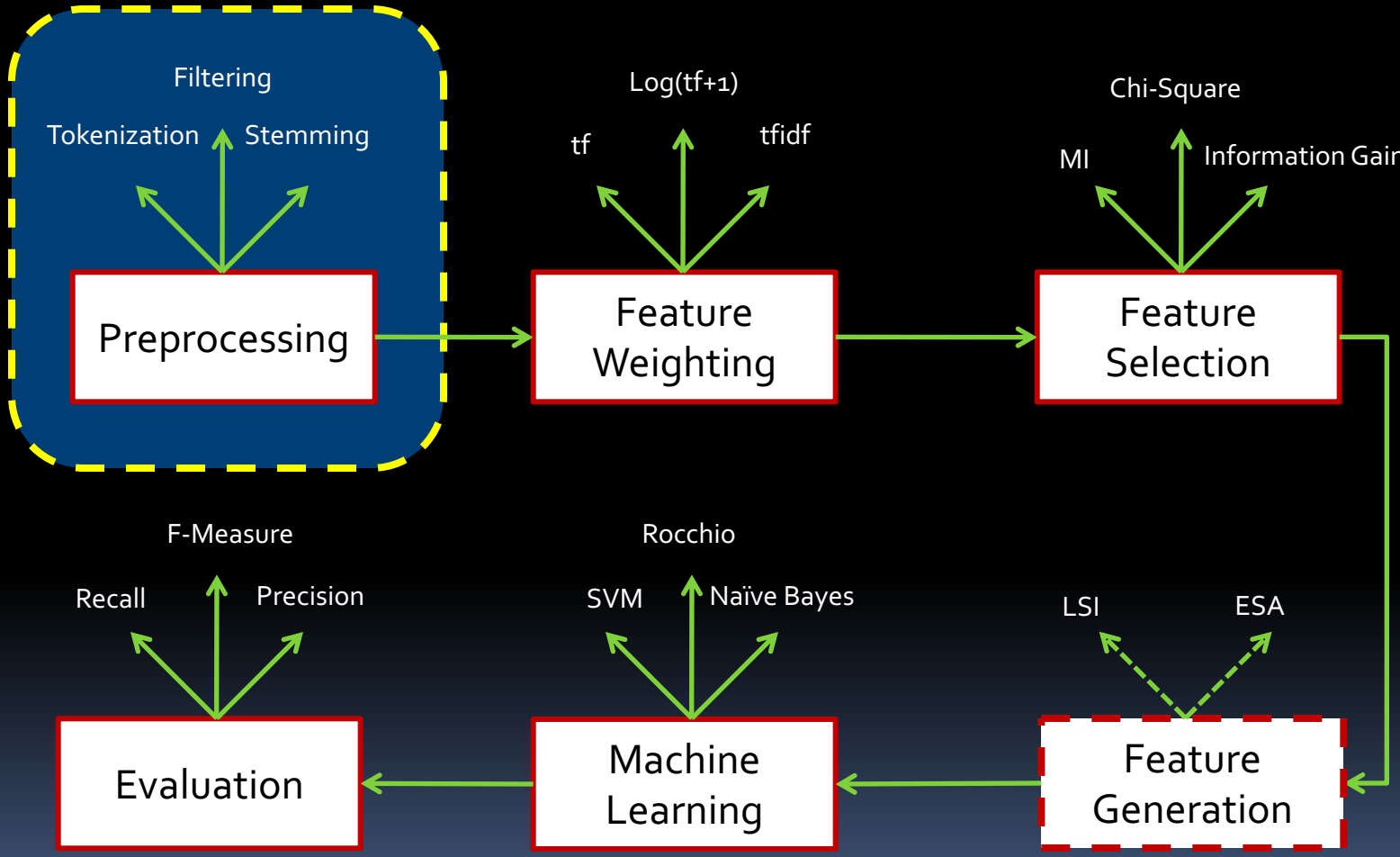
- Manual: Typically rule-based (KE Approach)
  - Does not scale up (labor-intensive, rule inconsistency)
  - May be appropriate for special data on a particular domain
- Automatic: Typically exploiting machine learning techniques
  - Vector space model based
    - Prototype-based (Rocchio)
    - **K-nearest neighbor (KNN)**
    - Decision-tree (learn rules)
    - Neural Networks (learn non-linear classifier)
    - **Support Vector Machines (SVM)**
  - Probabilistic or generative model based
    - **Naïve Bayes classifier**

Document Representation

- The idea is to process the natural language text in a document and transform into a vector



Anatomy of a Text Categorizer



Feature types

What will be your terms?

Most crucial decision you'll make!

1. Topic
   - **Words**, phrases (e.g. n-gram words)
2. Author
   - Stylistic features
3. Sentiment
   - Adjectives, ?
4. Spam
   - Specialized vocabulary

Terms

- Sophisticated Approaches (LSA, LDA, …)
- Higher Order statistics (?)
- Phrases (how to define?)
  - Syntactically – according to grammer (Noun phrases)
  - Statistically – strongly occurring patterns of words (Collocations?, n-gram words?)

Terms

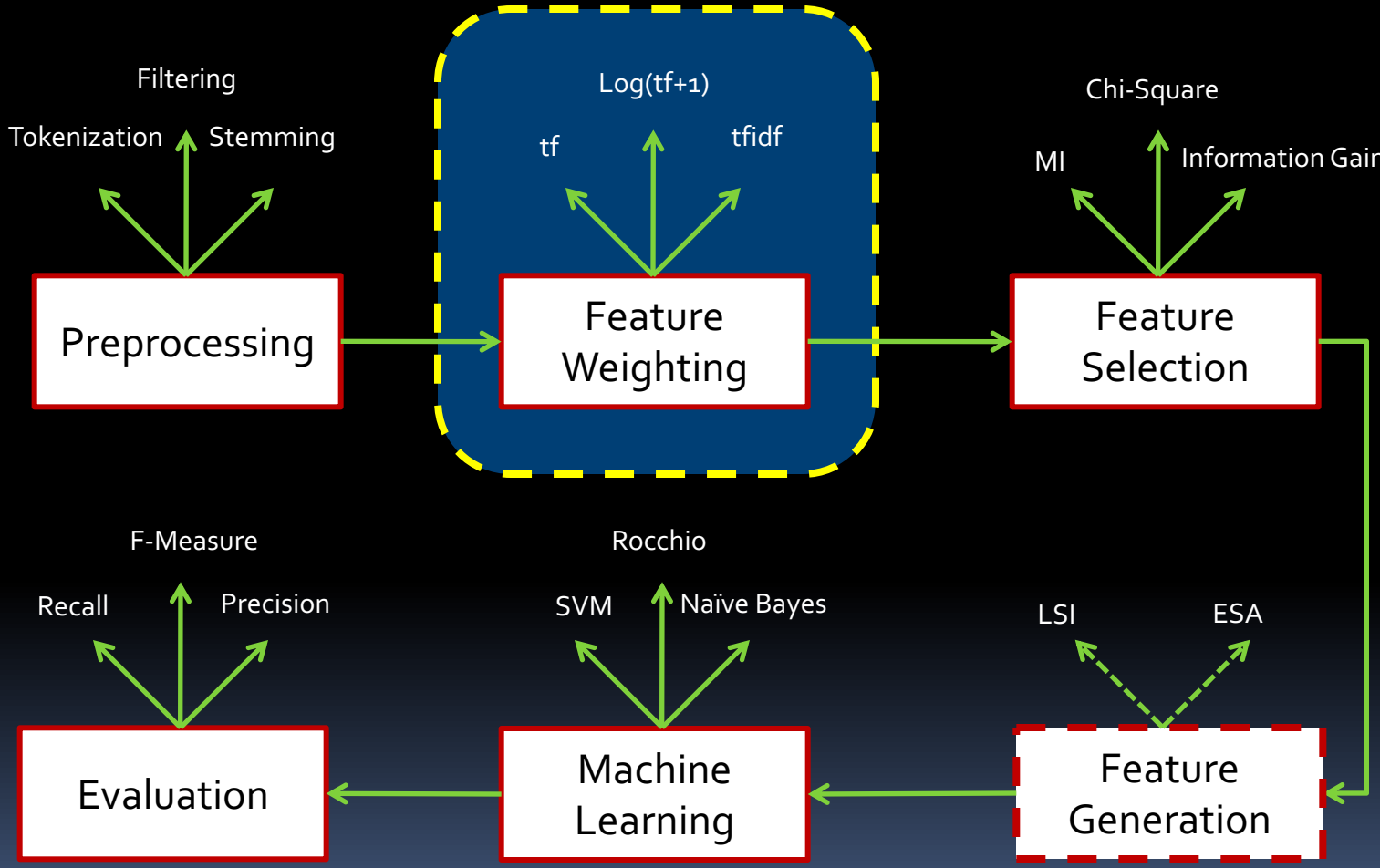
- Simplest approach: a term is a word
  - Bag of Words approach
- Preprocessing
  - Stopword removal (“a”,”the”,”of”,”and”)
  - Stemming (“stemming”,”stemmed”,”stemmer”)
  - Filtering
- Ignore word order

Filtering

- Simple filtering methods;
  - Pruning of infrequent words
    - Words are only considered as features, if they occur at least a few times in the training data.
    - **DF Filtering**
  - Pruning of high frequency words
    - This technique is supposed to eliminate non content words like "the", "and", "for".
    - **Stopword filtering**

A probabilistic Analysis of the Rocchio Algorithm with TFIDF for Text Categorization, Thorsten Joachims, Carnegie Mellon University, Pittsburgh, PA

Anatomy of a Text Categorizer



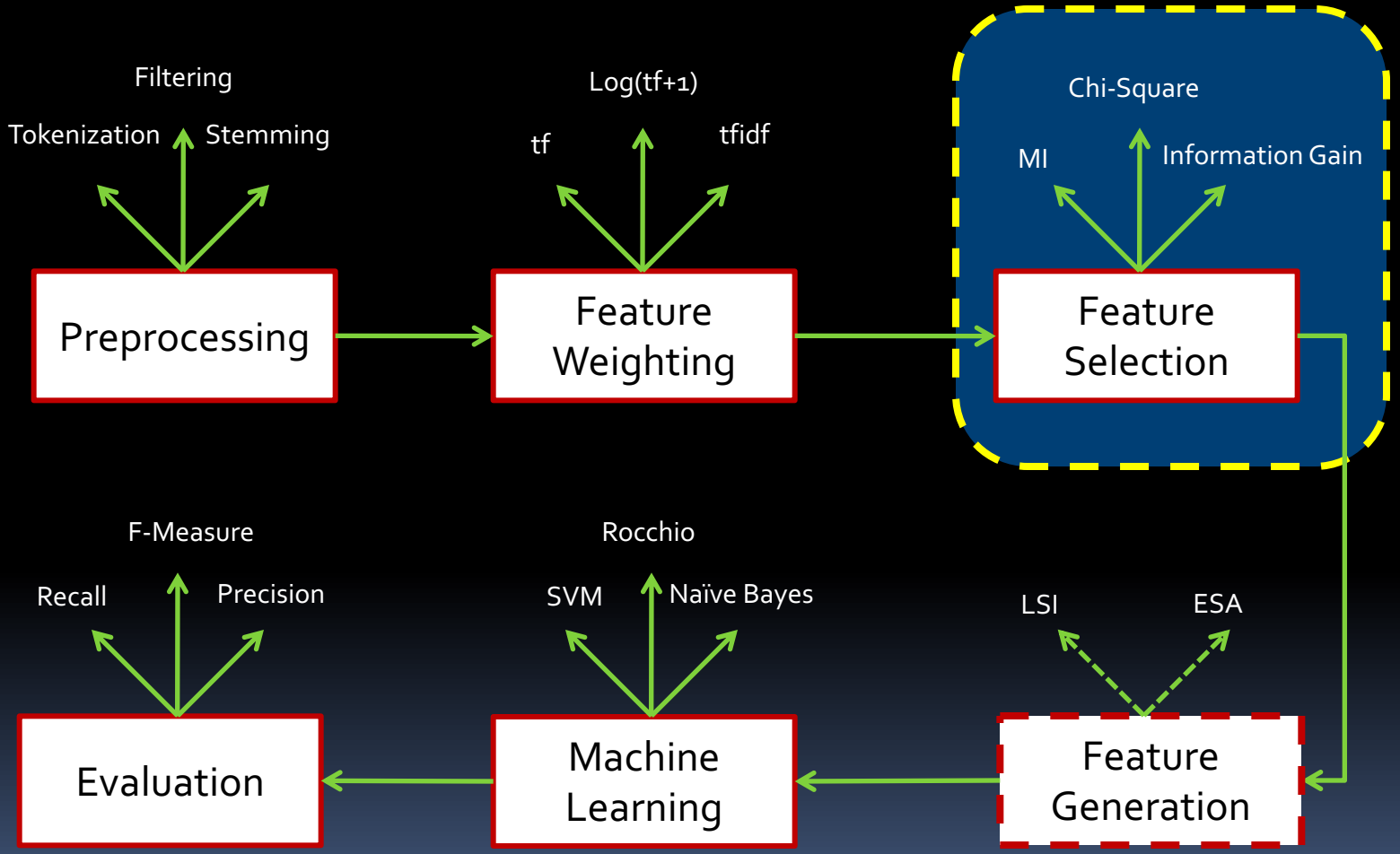
Weights

- Term frequency
- tf.idf $$tf.idf(t_k) = tf(t_k).idf(t_k)$$
  - The more often the term appears in a document, the more the representative is it of the document.
  - The more documents the term appears in the less discriminating it is.

$$idf(t_k) = \log(\frac{N_i}{N_k})$$

- Normalized tfidf $$w(t_k) = \frac{tf.idf(t_k)}{\sqrt{\sum_{s=1} (tf.idf(t_s))^2}}$$
  - Normalize the tf.idf values to the range[0,1]

Anatomy of a Text Categorizer



High Dimensionality

- There are many terms
- Many learning algorithms don't deal with extremely high dimensions
- Over fitting problem
- Not all terms are equally effective
- Solution? Eliminate unwanted terms

Feature Selection

- Also known as Dimensionality Reduction
- Idea: find a more efficient document representation, with much fewer dimensions, with a minimal loss of effectiveness (accuracy).
- Local vs. Global Policies
  - Local Policy: For each category, find the best terms.
  - Global Policy: Given all the categories find the best terms.

Feature Selection

- Out of original set of terms, t, find a much smaller subset, t', that yields high-test effectiveness(accuracy).
- Examples
  - Chi Square
  - Mutual Information
  - Information Gain
  - Information Ratio
  - Odd Ratio

Mutual Information

- Measure the association between to objects
  - It is a ratio of how many times the objects observed together normalized by the product of the occurrence of each object.

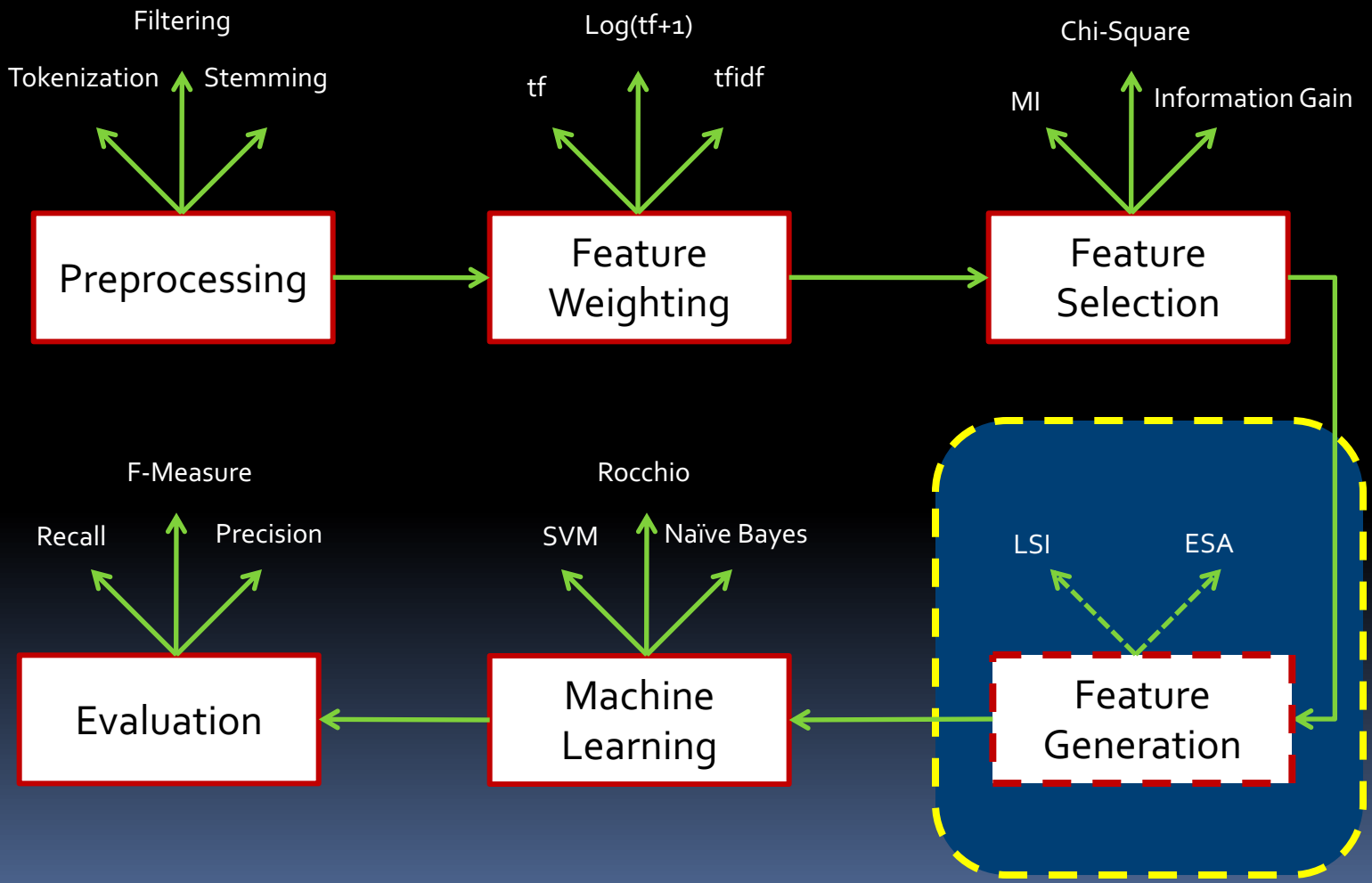
$$MI(A,B) = \frac{P(AB)}{P(A)P(B)}$$

Chi-Square

- The key idea of the chi-square test is a comparison of observed and expected values.

$$\chi^2 = \sum_{ij} \frac{(Observed_{ij} - Expected_{ij})^2}{Expected_{ij}}$$

Anatomy of a Text Categorizer



Feature Generation

- Term Clustering
  - Unsupervised
  - Supervised
  - Distributional clustering
- Latent Semantic Indexing
- Explicit Semantic Indexing

Latent Semantic Indexing (LSI)

- Words by themselves are not a good measure.
  - Synonyms (car, automobile)
  - Polysemous (Apple, Jaguar)
- LSI: a method for inferring the contextual similarity of terms
  - Finds the best m uncorrelated terms that best describe the original n terms.
  - Uncover latent information (synonyms)

Explicit Semantic Analysis

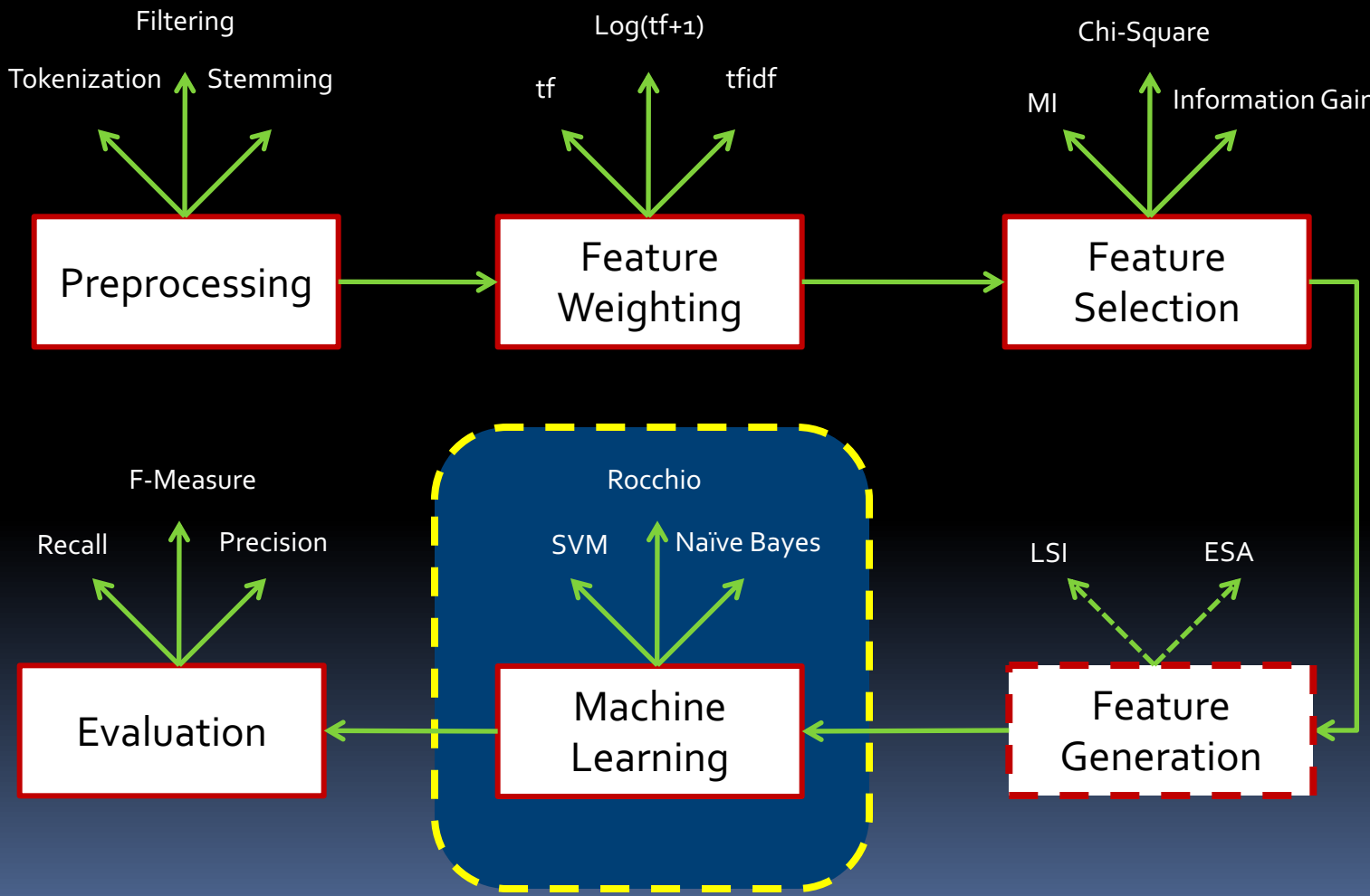
- Expand the terms using concept space (e.g. Wikipedia)
  - BOW

American politics → Democrats, Republicans, abortion, taxes, homosexuality, guns, etc

  - ESA

Car → Wikipedia:Car, Wikipedia:Automobile, Wikipedia:BMW, Wikipedia:Railway, etc

Anatomy of a Text Categorizer



Classification Methods

- Common classifiers in Text Classification
- Vector space model based
  - **K-nearest neighbor (KNN)**
  - **Support Vector Machines (SVM)**
- Probabilistic or generative model based
  - **Naïve Bayes classifier (NB)**

Naïve Bayes

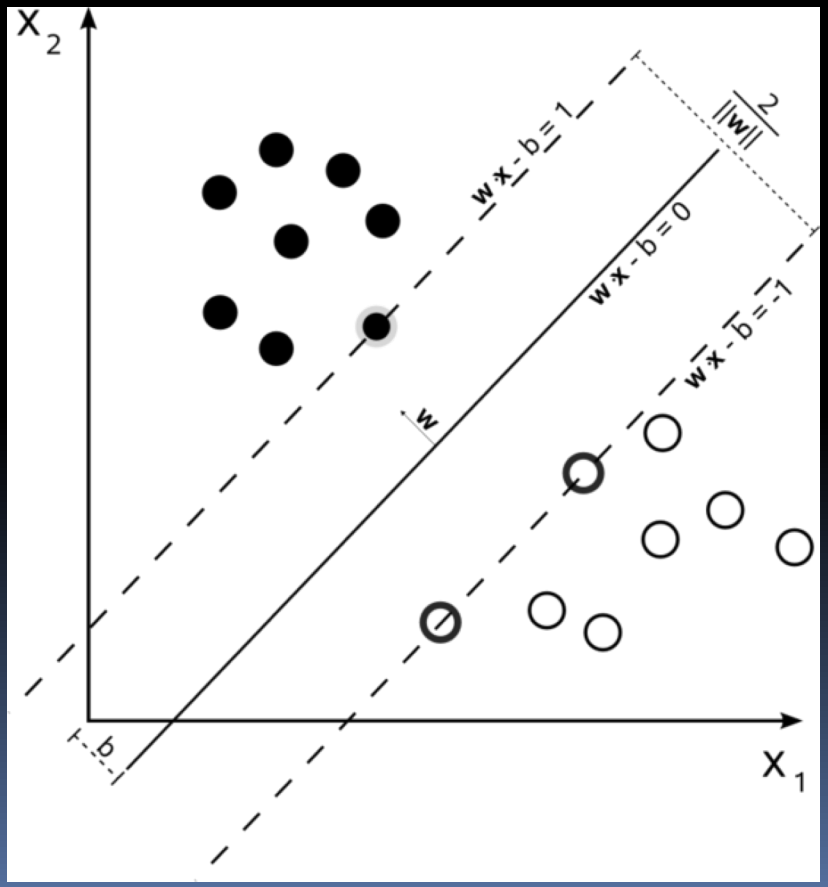
- Calculate the $P(C_k \mid D)$ the probability that document belong to the class C_k
- By Bayes' theorem

$$P(C_k \mid D) = \frac{P(C_k)P(D \mid C_k)}{P(D)}$$

$$P(D \mid C_k) = \prod_i P(w_i \mid C_k)$$

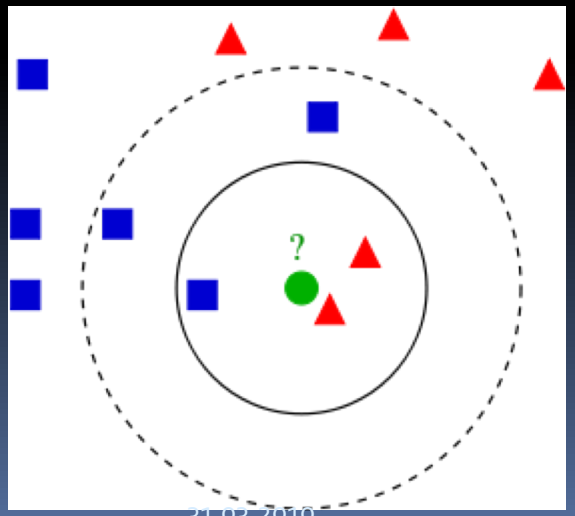
SVM

- Find the best hyper plan that separates the data points of two classes which a maximum separation (margin)

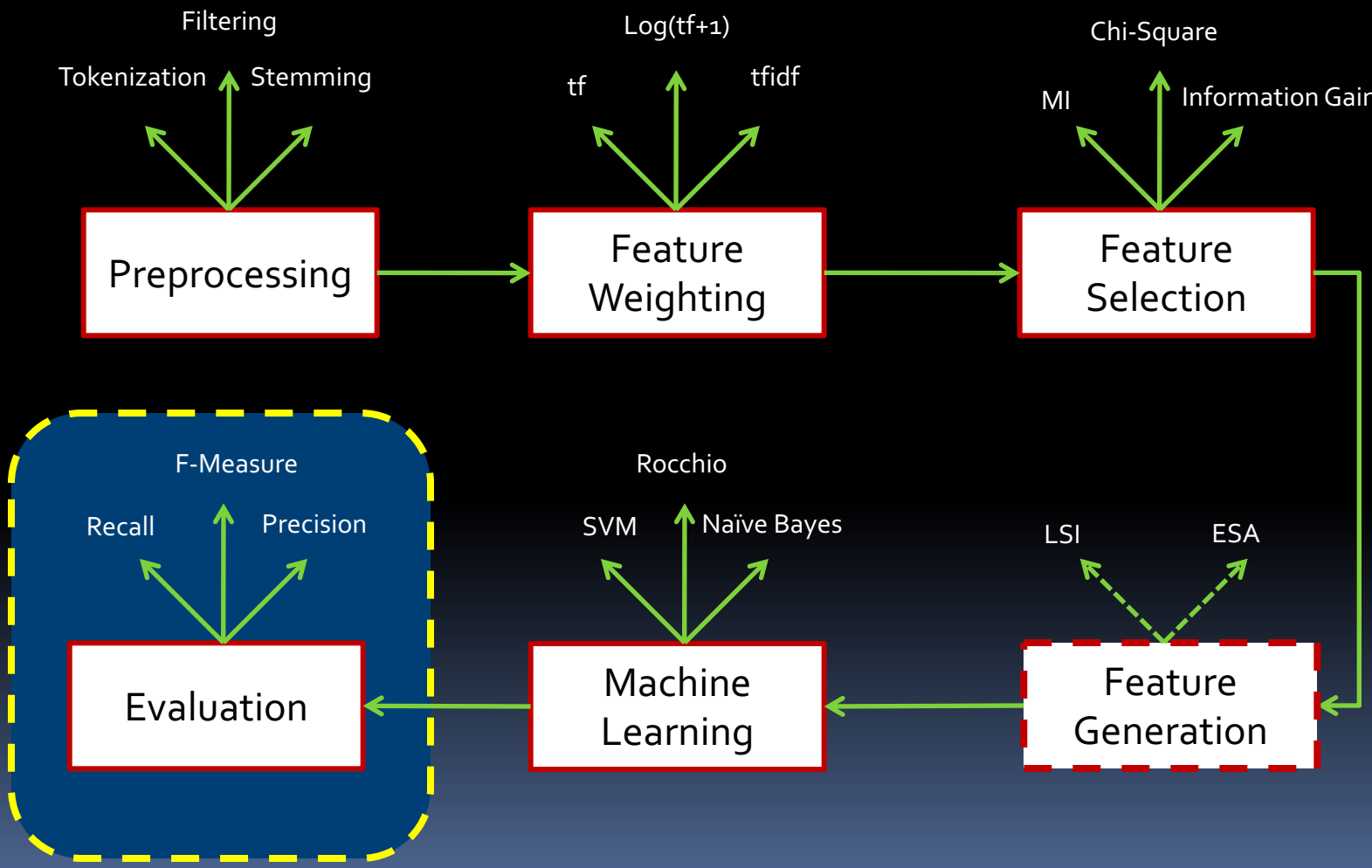


K Nearest Neighbor(K-NN)

- Document is classified by a majority vote of its neighbors, with the object being assigned to the class most common amongst its *k* nearest neighbors
- To measure the distance two vectors
  - Euclidian Distance
  - Cosine Angle



Anatomy of a Text Categorizer



Results & Evaluation

- How to measure the effectiveness of your classification?
  - Precision
  - Recall
  - F-Measure
  - Accuracy
  - Micro/Macro Averaging
  - Breakeven

Evaluations

- Effectiveness measure
  - Classic: Precision & Recall

Table II. The Contingency Table for Category c_i

Category c_i		Expert judgments	
		YES	**NO**
Classifier Judgments	**YES**	TP_i	FP_i
	NO	FN_i	TN_i

- Precision $\hat{\pi}_i = \frac{TP_i}{TP_i + FP_i}$
- Recall $\hat{\rho}_i = \frac{TP_i}{TP_i + FN_i}$.

Results & Evaluation

	Correct=Y	Correct=N
Assigned=Y	**a**	**b**
Assigned=N	**c**	**d**

Accuracy = (a+d)/(a+b+c+d)

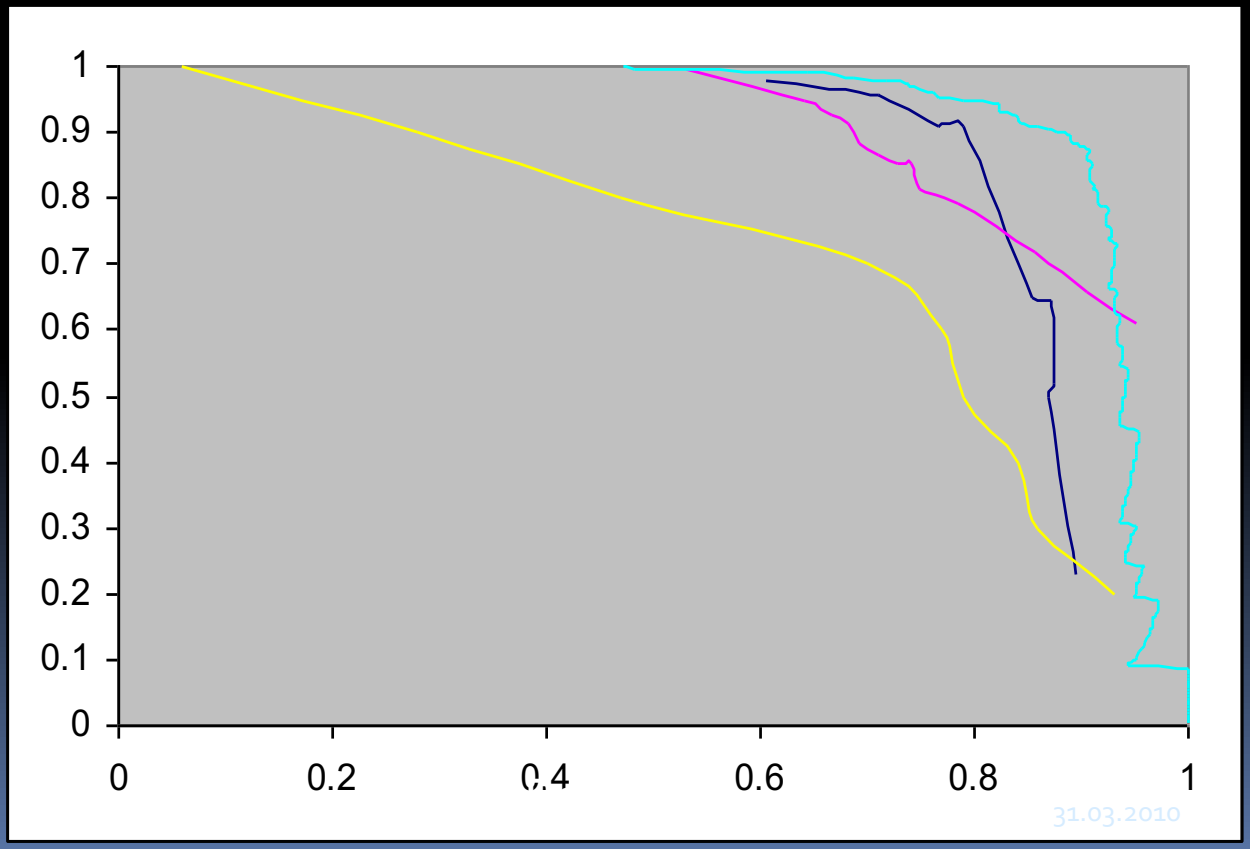
Precision = a/(a+b)

Recall= a/(a+c)

- F-Measure = 2*Precision*Recall/(Precision+Recall)
- Micro/Macro Averaging
- Breakeven (When Precision=Recall)

Breakeven point

breakeven point is the value where precision equals recall



Multiclass Problem: Micro- vs. Macro-Averaging

- If we have more than one class, how do we combine multiple performance measures into one quantity?
- Macroaveraging: Compute performance for each class, then average.
- Microaveraging: Collect decisions for all classes, compute contingency table, evaluate.

Micro- vs. Macro-Averaging: Example

Class 1

	Truth: yes	Truth: no
Classifier: yes	10	10
Classifier: no	10	970

Class 2

	Truth: yes	Truth: no
Classifier: yes	90	10
Classifier: no	10	890

Micro.Av. Table

	Truth: yes	Truth: no
Classifier: yes	100	20
Classifier: no	20	1860

- Macroaveraged precision: (0.5 + 0.9)/2 = 0.70
- Microaveraged precision: 100/120 = 0.83
- Why this difference?

Evaluation (con't)

- Benchmarks
  - Classic: Reuters collection
    - A set of newswire stories classified under categories related to economics
- Effectiveness
  - Difficulties of strict comparison
    - different parameter setting
    - different "split" (or selection) between training and testing
    - various optimizations … …
  - However, widely recognizable
    - Best: Boosting-based committee classifier & SVM
    - Worst: Naïve Bayes classifier
  - Need to consider other factors, especially efficiency

Example

Cats	Dogs
Cat ate your tongue?	My dog name is rocky.
I have a tuxedo cat.	Dogs are very affectionate.
Cats are relatives of tigers.	Dogs are descendents of wolves.

Preprocessing

Cats	Dogs
cat eat your tongue?	my dog name be rocky.
i have a tuxedo cat.	dog be very affectionate.
cat be relative of tiger.	dog be descendant of wolf.

Example

Cats	Dogs
cat eat your tongue?	my dog name is rocky.
i have a tuxedo cat.	dog be very affectionate.
cat be relative of tiger.	dog be descendant of wolf.

Cats	
Term	freq
cat	3
tuxedo	1
relative	1
tiger	1

Dogs	
Term	freq
dog	3
affectionate	1
rocky	1
wolf	1

Example

Cats		
Term	freq	P
cat	3	4/12
tuxedo	1	2/12
relative	1	2/12
tiger	1	2/12

Dogs		
Term	freq	P
dog	3	4/11
affectionate	1	2/11
rocky	1	2/11
wolf	1	2/11

The Toyger is an exciting new breed of domestic cats.

the toyger is an exciting new breed of domestic cat.

$$P(D \mid C_k) = \prod_i P(w_i \mid C_k)$$

$$P(C_k \mid D) \ \alpha \ P(C_k)P(D \mid C_k)$$

$$P(D \mid Cats) = P(toyger \mid Cats) * P(exciting \mid Cats) * P(breed \mid Cats) * P(cat \mid Cats)$$

$$P(D \mid Cats) = \frac{1}{12} * \frac{1}{12} * \frac{1}{12} * \frac{4}{12} = 1.9 *^{-4}$$

$$P(Cats \mid D) \ \alpha \ P(Cats)P(D \mid Cats) = \frac{3}{6} * (1.9 * 10^{-4}) = 9.5 * 10^{-5}$$

$$P(D \mid Dogs) = P(toyger \mid Dogs) * P(exciting \mid Dogs) * P(breed \mid Dogs) * P(cat \mid Dogs)$$

$$P(D \mid Dogs) = \frac{1}{11} * \frac{1}{11} * \frac{1}{11} * \frac{1}{11} = 6.8 * 10^{-5}$$

$$P(Dogs \mid D) \ \alpha \ P(Dogs)P(D \mid Dogs) = \frac{3}{6} * (6.8 * 10^{-5}) = 3.4 * 10^{-5}$$

Questions?

Next Experiments & Results from Literature…

Reuters-21578: Typical document

<REUTERS TOPICS="YES" LEWISSPLIT="TRAIN" CGISPLIT="TRAINING-SET" OLDID="12981" NEWID="798">

<DATE> 2-MAR-1987 16:51:43.42</DATE>

<TOPICS><D>livestock</D><D>hog</D></TOPICS>

<TITLE>AMERICAN PORK CONGRESS KICKS OFF TOMORROW</TITLE>

<DATELINE> CHICAGO, March 2 - </DATELINE><BODY>The American Pork Congress kicks off

tomorrow, March 3, in Indianapolis with 160 of the nations pork producers from 44 member states determining industry positions on a number of issues, according to the National Pork Producers Council, NPPC.

Delegates to the three day Congress will be considering 26 resolutions concerning various issues, including the future direction of farm policy and the tax law as it applies to the agriculture sector. The delegates will also debate whether to endorse concepts of a national PRV (pseudorabies virus) control and eradication program, the NPPC said.

A large trade show, in conjunction with the congress, will feature the latest in technology in all areas of the industry, the NPPC added. Reuter

&#3;</BODY></TEXT></REUTERS>

Reuters 21578

- Most (over)used data set (c. 1998)
- 21578 documents
- Average document length: 200 words
- 9603 training, 3299 test articles (ModApte split)
- 118 categories
- article can be in > 1 category (average: 1.24)
- only about 10 out of 118 categories are large

Common categories (#train, #test)

- Earn (2877, 1087)
- Acquisitions (1650, 179)
- Money-fx (538, 179)
- Grain (433, 149)
- Crude (389, 189)
- Trade (369,119)
- Interest (347, 131)
- Ship (197, 89)
- Wheat (212, 71)
- Corn (182, 56)

First Experiment: Yang and Liu

- Features: stemmed words (stop words removed)
- Indexing: frequency (?)
- Feature filtering: top infogain words (1000 to 10000)
- Evaluation: macro- and micro-averaged F1

Results: Yang&Liu

Table 1: Performance summary of classifiers

method	miR	miP	miF1	maF1	error
SVM	.8120	.9137	.8599	.5251	.00365
KNN	.8339	.8807	.8567	.5242	.00385
LSF	.8507	.8489	.8498	.5008	.00414
NNet	.7842	.8785	.8287	.3765	.00447
NB	.7688	.8245	.7956	.3886	.00544

miR = micro-avg recall; miP = micro-avg prec.;
miF1 = micro-avg F1; maF1 = macro-avg F1.

Second Experiment: Dumais et al

- Features: non-rare words
- Indexing: binary
- Feature filtering: top infogain words (30 per category)
- Evaluation: macro-averaged break-even

Results: Dumais et al. Breakeven

	Rocchio	NBayes	C4.5	LinearSVM	
earn	92.9%	95.9%	97.8%	98.2%	
acq	64.7%	87.8%	89.7%	92.8%	
money-fx	46.7%	56.6%	66.2%	74.0%	
grain	67.5%	78.8%	85.0%	92.4%	
crude	70.1%	79.5%	85.0%	88.3%	
trade	65.1%	63.9%	72.5%	73.5%	
interest	63.4%	64.9%	67.1%	76.3%	
ship	49.2%	85.4%	74.2%	78.0%	
wheat	68.9%	69.7%	92.5%	89.7%	
corn	48.2%	65.3%	91.8%	91.1%	
Avg Top 10	64.6%	81.5%	88.4%	91.4%	
Avg All Cat	61.7%	75.2%	na	86.4%	

Observations: Dumais et al

- Features: words + bigrams

  No improvement!

- Indexing: frequency instead of binary

  No improvement!

Third Experiment: Joachims

- Features: stemmed unigrams (stop words removed)
- Indexing: tf*idf
- Feature filtering: 1000 top infogain words
- Evaluation: micro-averaged break-even

Results: Joachims

	Bayes	Rocchio	C4.5	k-NN	SVM (poly) degree $d=$ 1	2	3	4	5	SVM (rbf) width $\gamma=$ 0.6	0.8	1.0	1.2
earn	95.9	96.1	96.1	97.3	98.2	98.4	**98.5**	98.4	98.3	**98.5**	98.5	98.4	98.3
acq	91.5	92.1	85.3	92.0	92.6	94.6	**95.2**	95.2	95.3	95.0	95.3	95.3	**95.4**
money-fx	62.9	67.6	69.4	78.2	66.9	72.5	75.4	74.9	**76.2**	74.0	75.4	**76.3**	75.9
grain	72.5	79.5	89.1	82.2	91.3	93.1	**92.4**	91.3	89.9	**93.1**	91.9	91.9	90.6
crude	81.0	81.5	75.5	85.7	86.0	87.3	88.6	**88.9**	87.8	**88.9**	89.0	88.9	88.2
trade	50.0	77.4	59.2	77.4	69.2	75.5	76.6	77.3	**77.1**	76.9	78.0	**77.8**	76.8
interest	58.0	72.5	49.1	74.0	69.8	63.3	67.9	73.1	**76.2**	74.4	75.0	**76.2**	76.1
ship	78.7	83.1	80.9	79.2	82.0	85.4	86.0	**86.5**	86.0	**85.4**	86.5	87.6	87.1
wheat	60.6	79.4	85.5	76.6	83.1	84.5	85.2	**85.9**	83.8	**85.2**	85.9	85.9	85.9
corn	47.3	62.2	87.7	77.9	86.0	86.5	85.3	**85.7**	83.9	**85.1**	85.7	85.7	84.5
microavg.	**72.0**	**79.9**	**79.4**	**82.3**	84.2	85.1	85.9	86.2	85.9	86.4	86.5	86.3	86.2
					combined: **86.0**					combined: **86.4**			

Questions?

Next Clustering…

Document Clustering

- Motivation
  - Automatically group related documents based on their contents
  - No predetermined training sets or taxonomies
  - Generate a taxonomy at runtime
- Most popular clustering methods are:
  - K-Means clustering
  - Agglomerative hierarchical clustering
  - EM (Gaussian Mixture)
  - …

The Steps and Algorithms

- Clustering Process
  - Data preprocessing: remove stop words, stem, feature extraction, lexical analysis, etc.
  - Hierarchical clustering: compute similarities by applying clustering algorithms
  - Model-Based clustering (Neural Network Approach): clusters are represented by "exemplars" (e.g., SOM)

K-Means clustering

- Given:
  - set of documents (e.g., TFIDF vectors),
  - distance measure (e.g., cosine)
  - K (number of groups)
- For each of K groups, initialize its centroid with a random document
- While not converging
  - Each document is assigned to the nearest group (represented by its centroid)
  - For each group, calculate new centroid (group mass point, average document in the group)

K-means

1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to.
4. Each Center finds the centroid of the points it owns…
5. …and jumps there
6. …Repeat until terminated!

